

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 1: EXPLORATION: ENTERING THE WORLD OF SECONDARY SCIENCE [SET B - ADVANCED HOTS]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH01/SET-B****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET B - HIGHER ORDER THINKING SKILLS):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Advanced Conceptual and Numerical MCQs with single correct option.
3. **Section B (Q16 – Q20):** Critical Assertion-Reasoning based questions testing analytical clarity.
4. **Section C (Q21 – Q25):** Case-Study / Practical Experimental Data-interpretation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: ADVANCED & NUMERICAL MCQS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 The main scale of a vernier caliper reads in millimeters. If 10 vernier scale divisions coincide with 9 main scale divisions, what is the least count of the instrument?

- (A) 0.1 mm (0.01 cm) (B) 1.0 mm (0.1 cm)
(C) 0.01 mm (0.001 cm) (D) 0.9 mm (0.09 cm)

Q.2 When the jaws of a screw gauge are brought into contact, the zero of the circular scale lies 3 divisions below the reference datum line. This zero error is:

- (A) Positive zero error (must be subtracted from observed reading) (B) Negative zero error (must be added to observed reading)
(C) Random parallax error (D) Instrumental backlash error

Q.3 A student measures the mass of an object as 25.0 g and its volume as 10.0 cm³. What is the density expressed with the correct number of significant figures?

- (A) 2.5 g/cm³ (B) 2.50 g/cm³
(C) 2.500 g/cm³ (D) 0.40 g/cm³

Q.4 Which of the following is equivalent to 1 kilogram per cubic meter (1 kg/m^3) in CGS units?

- (A) 1000 g/cm^3 (B) 10^{-3} g/cm^3
(C) 100 g/cm^3 (D) 10^{-6} g/cm^3

Q.5 A student measures a wire's diameter as 2.4 mm, 2.5 mm, 2.4 mm, and 2.7 mm. The mean diameter of the wire is:

- (A) 2.5 mm (B) 2.4 mm
(C) 2.6 mm (D) 2.45 mm

Q.6 When heating an inflammable liquid such as ethanol in a chemistry laboratory, which apparatus should always be employed to avoid fire hazard?

- (A) Direct heating over an open Bunsen flame
(B) A thermostatically controlled water bath
(C) A blowpipe flame
(D) Direct heating in an open silica crucible

Q.7 An irregular stone of mass 78 g is lowered into a measuring cylinder containing 45 mL of water. The water level rises to 55 mL. What is the density of the stone?

- (A) 1.42 g/cm³
(B) 7.8 g/cm³
(C) 0.78 g/cm³
(D) 780 g/cm³

Q.8 How many millimeters are there in 3.5×10^{-2} kilometers?

- (A) 35,000 mm
(B) 3,500 mm
(C) 350 mm
(D) 350,000 mm

Q.9 In an experiment plotting rate of gas production against temperature, what is the best curve fit when data points show slight experimental scatter?

- (A) Connecting every individual point with zig-zag sharp lines
(B) Drawing a smooth line/curve of best fit that balances points on either side
(C) Ignoring all points except the first and last
(D) Drawing a horizontal line through the origin

Q.10 A piece of copper of mass 89 g is combined with 178 g of another metal of density 8.9 g/cm³ to form a uniform alloy. If copper density is 8.9 g/cm³, what is the total volume of the alloy?

- (A) 10 cm³
(B) 20 cm³
(C) 30 cm³
(D) 40 cm³

Q.11 Which SI base unit is defined by fixing the numerical value of the Planck constant h to exactly $6.62607015 \times 10^{-34}$ in units of J s?

- (A) Meter
(B) Second
(C) Kilogram
(D) Ampere

Q.12 When observing mercury in a glass tube, the meniscus curves upward (convex). The correct reading is taken at:

- (A) The lowest point of the meniscus
(B) The highest central point (crest) of the meniscus
(C) The boundary touching the glass wall
(D) The mid-point of the meniscus curve

Q.13 A stopwatch has a smallest division of 0.2 s. A runner's time is recorded as 12.4 s. The percentage uncertainty in this single reading is approximately:

- (A) 1.6%
(B) 0.2%
(C) 5.0%
(D) 0.8%

Q.14 Which of the following variables must be manipulated to test if plant height depends on light intensity?

- (A) Keep light intensity constant while changing soil type
(B) Vary light intensity while keeping soil, water, and temperature identical
(C) Vary water, fertilizer, and light intensity simultaneously
(D) Keep all plants in complete darkness

Q.15 Which hazardous chemical symbol displays a skull and crossbones inside a red diamond frame?

- (A) Flammable substance
(B) Acute severe toxicity / Poison
(C) Corrosive to metals and skin
(D) Explosive material

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): Systematic errors cannot be completely eliminated by taking the arithmetic average of a large number of readings.

Reason (R): Systematic errors consistently skew measurements in one specific direction (consistently too high or consistently too low).

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): A hypothesis must be capable of being disproven (falsifiable) to be considered scientific.

Reason (R): A statement that cannot be tested or refuted by any possible empirical observation lies outside the domain of science.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): Density is an intensive physical property of a homogeneous pure substance.

Reason (R): The ratio of mass to volume remains constant regardless of the total quantity of the substance taken.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): When smelling chemical vapors in a laboratory, one should inhale deeply with the nose directly over the open container.

Reason (R): Deep inhalation allows accurate sensory classification of chemical odors.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) Both (A) and (R) are false.

Q.20 Assertion (A): SI units are based on decimal multiples and sub-multiples.

Reason (R): Converting between units in the metric SI system simply involves multiplying or dividing by powers of ten.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL DATA & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Calibrating and Measuring Metal Alloy Density

An engineer wants to find the exact density of a newly fabricated metallic cylinder. Using a vernier caliper with a least count of 0.01 cm, she measures the diameter $\phi D = 2.00$ cm and height $H = 5.00$ cm. Using an electronic analytical balance, the mass of the cylinder is determined to be 125.60 g. Take $\pi = 3.14$.

Q.21 What is the radius r of the metallic cylinder in centimeters?

- (A) 1.00 cm (B) 2.00 cm
(C) 0.50 cm (D) 4.00 cm

Q.22 What is the calculated volume of the cylinder using $V = \pi r^2 H$?

- (A) 15.70 cm^3 (B) 31.40 cm^3
(C) 62.80 cm^3 (D) 7.85 cm^3

Q.23 What is the density of the metal cylinder calculated from the measurements?

- (A) 8.00 g/cm^3 (B) 4.00 g/cm^3
(C) 12.56 g/cm^3 (D) 1.57 g/cm^3

Q.24 If the metal cylinder is made of stainless steel (density 8.00 g/cm^3), what will be its density expressed in SI units (kg/m^3)?

- (A) 800 kg/m^3 (B) $8,000 \text{ kg/m}^3$
(C) $80,000 \text{ kg/m}^3$ (D) 80 kg/m^3

Q.25 If the electronic balance had a zero error of $+0.60 \text{ g}$ that was uncorrected, what was the true mass and true density of the cylinder?

- (A) 126.20 g and 8.04 g/cm^3 (B) 125.00 g and 7.96 g/cm^3
(C) 125.00 g and 8.00 g/cm^3 (D) 124.40 g and 7.90 g/cm^3

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET B)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & IN-DEPTH EXPLANATIONS (SET B)

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.1	(A)	Least Count $= 1 \times \text{MSD} - 1 \times \text{VSD} = 1 \times \text{mm} - 0.9 \times \text{mm} = 0.1 \times \text{mm} = 0.01 \times \text{cm}$.
Q.2	(A)	When the zero of the circular scale lies below the reference line, the gauge reads positive without load. Hence it is positive zero error and must be subtracted from the final reading.
Q.3	(B)	Both 25.0 g and 10.0 cm^3 have 3 significant figures. The quotient $25.0 / 10.0 = 2.50 \text{ g/cm}^3$ must also have 3 significant figures.
Q.4	(B)	$1 \text{ kg/m}^3 = \frac{1000 \text{ g}}{(100 \text{ cm})^3} = \frac{10^3}{10^6} \text{ g/cm}^3 = 10^{-3} \text{ g/cm}^3$ (or 0.001 g/cm^3).
Q.5	(A)	$\text{Mean} = \frac{2.4 + 2.5 + 2.4 + 2.7}{4} = \frac{10.0}{4} = 2.5 \text{ mm}$.
Q.6	(B)	Ethanol is highly inflammable (boils at 78°C). Its vapors can ignite easily; hence it must be heated inside a boiling water bath, never directly over an open flame.
Q.7	(B)	$\text{Displaced volume} = 55 - 45 = 10 \text{ mL} = 10 \text{ cm}^3$. $\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{78 \text{ g}}{10 \text{ cm}^3} = 7.8 \text{ g/cm}^3$.
Q.8	(A)	$1 \text{ km} = 10^6 \text{ mm}$. Therefore, $3.5 \times 10^{-2} \times 10^6 = 3.5 \times 10^4 = 35,000 \text{ mm}$.
Q.9	(B)	A line or curve of best fit minimizes random observational errors by showing the overall mathematical trend rather than connecting random fluctuations.
Q.10	(C)	$V_{\text{Cu}} = 89 / 8.9 = 10 \text{ cm}^3$. $V_{\text{metal}} = 178 / 8.9 = 20 \text{ cm}^3$. $\text{Total volume} = 10 + 20 = 30 \text{ cm}^3$.
Q.11	(C)	Under the revised SI definitions (2019), the kilogram (kg) is defined based on the fundamental Planck constant h .
Q.12	(B)	Mercury does not wet glass (cohesive forces exceed adhesive forces), forming a convex meniscus. Hence, the reading is taken at the top/crest at eye level.
Q.13	(A)	$\text{Percentage uncertainty} = \frac{\text{Least count}}{\text{Measured value}} \times 100 = \frac{0.2}{12.4} \times 100 \approx 1.61\%$.
Q.14	(B)	A valid scientific test changes only the independent variable (light intensity) while keeping all other confounding variables strictly controlled.
Q.15	(B)	The Globally Harmonized System (GHS) skull and crossbones pictogram denotes acute severe toxicity (fatal or poisonous upon exposure).
Q.16	(A)	Averaging reduces random Gaussian errors, but systematic errors (like zero error or faulty calibration) always shift readings unidirectionally, so averaging cannot eliminate them.
Q.17	(A)	According to the scientific method (Karl Popper's criterion of falsifiability), scientific hypotheses must make testable predictions that could potentially fail in experiments.
Q.18	(A)	Intensive properties do not depend on system size or amount of material. Since doubling mass also doubles volume, the ratio (density) remains constant.
Q.19	(D)	Both statements are completely false and dangerous! Never inhale directly; gently waft vapors towards the nose using your hand from a safe distance.
Q.20	(A)	The metric SI system uses base 10 prefixes (kilo, mega, centi, milli, micro), making mathematical conversion straightforward without arbitrary conversion fractions.
Q.21	(A)	$\text{Radius } r = D / 2 = 2.00 / 2 = 1.00 \text{ cm}$.
Q.22	(A)	$V = \pi r^2 H = 3.14 \times (1.00)^2 \times 5.00 = 15.70 \text{ cm}^3$.

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.23	(A)	$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{125.60 \text{ g}}{15.70 \text{ cm}^3} = 8.00 \text{ g/cm}^3$.
Q.24	(B)	To convert from g/cm^3 to kg/m^3 , multiply by 1000: $8.00 \times 1000 = 8,000 \text{ kg/m}^3$.
Q.25	(B)	$\text{True mass} = 125.60 - 0.60 = 125.00 \text{ g}$. $\text{True density} = 125.00 / 15.70 \approx 7.96 \text{ g/cm}^3$.